10.1 Event-driven architectures

An event is an occurrence. It's something relevant that happened in a system, like a state change, and there can be many sources of events. This chapter will focus on applications, but events can very well be happening in IoT devices, sensors, or networks. When an event occurs, interested parties can be notified. Event notification is usually done through messages, which are data representations of events.

In an event-driven architecture, we identify *event producers* and *event consumers*. A producer is a component that detects the event and sends a notification. A consumer is a component that is notified when a specific event occurs. Producers and consumers don't know each other and work independently. A producer sends an event notification by publishing a message to a channel operated by an event broker that's responsible for collecting and routing messages to consumers. A consumer is notified by the broker when an event occurs and can act upon it.

Producers and consumers have minimal coupling when using a broker that takes the processing and distribution of events on itself. In particular, they are temporally decoupled, because the interaction is asynchronous. Consumers can fetch and process messages at any time without affecting the producers whatsoever.

In this section, you'll learn the fundamentals of event-driven models and how they can help build more resilient and loosely coupled applications in the cloud.

10.1.1 Understanding the event-driven models

Event-driven architectures can be based on two main models: